κ_E is Prior Art: Reconciling the Structural Epistemic Cut with He & Yu (arXiv 2609.02925)

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`work-in-progress @ 0bbabeb`
Corrects the min-cut κ_E in my S230 PDF `668cdb6`

1. Claim

In S230 I sent you a vertex-minimum-cut (Menger) formulation of an "epistemic cut" κ_E for judge/quorum co-failure. He & Yu, *The Illusion of Independent Quorums*, arXiv 2609.02925 (submitted 24 Aug 2026), already coined the "Structural Epistemic Cut κ_E" and proved the headline theorem roughly two months before my PDF. The term and the structural result are prior art. We should cite them; we should not claim κ_E as ours.

2. What He & Yu have

- κ_E as a minimum set-cover over an explicit Epistemic Fault Basis $\mathcal{B}$ (their Definition 7): the minimum number of modelled root faults whose exposures collectively cover a minimal decisive coalition.
- Theorem 2: for every $q \geq 1$ and every non-trivial monotone approval rule there is an admissible exposure map with $\kappa_E = 1$; hence $|Q| \to \infty$ does *not* force $\kappa_E \to \infty$ (arbitrarily large quorums, one shared dependency).
- Corollary 1: adding voters at a fixed threshold cannot increase the cut.
- Theorem 1: κ_E lower-bounds the number of roots required for semantic compromise κ_S (under closed causal accounting, conservative exposure, authorisation alignment).
- A Dependency-Aware Quorum Controller (DAQC): a static, per-decision admission gate. A frozen 120-task benchmark validating the structural mechanism.
- Slogan: "replication does not imply epistemic redundancy."

3. How my min-cut differs — and why it does not rescue novelty of the term

Theirs is a set-cover over a declared fault basis (NP-hard in general; the basis and exposure map are required inputs). Mine is a vertex minimum cut from the independent input roots to the decision node (Menger; polynomial-time once the dependency DAG is given). Minimum cut is a special case of the same covering/duality structure — Menger is LP-duality specialised to path structure — so the min-cut version is a cleaner, computable *reformalization* of their object, not

a new object. Honest caveat: I cannot certify my min-cut version was independent rather than absorbed from an earlier browse. It does not matter — they hold a public, timestamped priority on the term and the theorem.

4. What survives as ours — and the paper corroborates the gap

Both of our genuine contributions are territories their paper explicitly declines to enter:

- **(a) The $\kappa_E \to n_{\text{eff}}$ statistical bridge.** They deliberately keep structural and statistical dependence separate; κ_E is "a basis-relative structural property of a declared model, not a calibrated probability of semantic truth" (their §7). We turn the structural cut into a statistical effective number via Kish-on-$\bar{\varphi}$: structural κ_E is the *leading* indicator, measured $\bar{\varphi} \to n_{\text{eff}}$ the *lagging* one. Absent in theirs.
- **(b) Sequential, anytime-valid monitoring.** DAQC is a static per-decision gate; they list "dynamic cut admission under streaming telemetry" as future work, and there is no martingale / e-value / e-process anywhere in the paper. Our Leg-3 cross-item e-process *is* that anytime-valid monitor. Absent in theirs.

Net: 2609.02925 is strong corroboration — an independent formal team proved replication $\neq$ epistemic redundancy — that takes the name and vacates exactly our two contributions.

5. Proposal and question

I propose we (i) cite He & Yu as the origin of κ_E / Epistemic Fault Domains, (ii) reframe our contribution as the structural $\to$ statistical bridge and the anytime-valid monitor, and (iii) stop calling κ_E "ours." Two questions for you: does your reading of 2609.02925 match this? And is there any of their set-cover machinery (strictly more general than my min-cut) we should adopt rather than discard?